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CERTIFICATIONS: AWS Certified Solutions Architect – Associate

Summary

Senior Software Engineer with 7+ years of experience building backend and data platforms, with recent focus on Generative AI solutions. Strong in RAG architectures using LangChain, FAISS, Pinecone, and Hugging Face Transformers, integrating LLMs via OpenAI GPT-4, Anthropic Claude, and AWS Bedrock with multi-step workflows, evaluation, and guardrails. Experienced in Python microservices (FastAPI/Django), ETL pipelines (Pandas/NumPy/SQL), and cloud-native deployments on AWS (Lambda, Step Functions, Glue, Athena, EC2, S3, IAM, KMS, Secrets Manager, VPC) using Docker/Kubernetes (EKS) with production observability (CloudWatch, Prometheus, Grafana, ELK) and secure access controls in Agile/Scrum environments.

Technical skills

Category	Technologies / Tools
Core AI Skills	LLMs, Generative AI, NLP, Deep Learning, RAG, Prompt Engineering, AI/GenAI Feature Integration
Programming Languages	Python, SQL, JavaScript, TypeScript, HTML/CSS, Shell Scripting (Bash), AsyncIO / Asynchronous Python, JSP, XML
AI / ML Frameworks	PyTorch, TensorFlow, Hugging Face Transformers
Data & Pipelines	ETL, Pandas, NumPy, Batch Processing, Asynchronous Processing
APIs & Backend	REST APIs, FastAPI, Flask, Django, Microservices Architecture
Databases	MySQL, MongoDB, PostgreSQL, DynamoDB, Redis, Elasticsearch
AWS Cloud services	Lambda, Glue, Step Functions, API Gateway, DynamoDB, SQS, Event Bridge, CloudFormation, S3, EC2, RDS, IAM, KMS, Secrets Manager, VPC, CloudTrail
DevOps & Deployment	Docker, Kubernetes (AWS EKS), Terraform, Jenkins, GitLab CI/CD, GitHub Actions, ArgoCD, AWS Code Pipeline
MLOps Tools	MLflow, Sage Maker Processing / Batch Inference, Monitoring, Drift Detection
Version Control	Git, GitHub, Bitbucket
Monitoring&Logging	Prometheus, Grafana, CloudWatch, Open Telemetry, ELK Stack, AWS X-Ray, Logstash, Kibana
Security & Compliance	IAM, KMS, Secrets Manager, RBAC, Encryption, GDPR, PII Compliance, OAuth2, JWT
Methodologies	Agile, Scrum, Continuous Integration, Continuous Delivery, DevSecOps, Code Reviews, Architecture Design Reviews

Professional Experience

Software Engineer

February 2024-Present

Morgan Stanley, US

Developed an internal GenAI assistant by building Python/FastAPI services that enabled operations teams to search policies and SOPs and receive grounded answers with citations. Implemented a full RAG workflow with LangChain, document ingestion, embeddings, and multi-step agent logic to improve response reliability, while supporting scalable deployments and performance through AWS services and caching. Ensured production readiness by deploying on Kubernetes, implementing observability and security controls, and adding automated tests for end-to-end validation.

- Built a **Python and FastAPI** backend for an internal **GenAI assistant** used by operations teams to search policies/SOPs and answer workflow questions with source citations.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain, including document chunking and metadata tagging; indexed PDF and structured documents using FAISS (dev/local) and Pinecone (staging/production).
- Integrated **OpenAI GPT-4**, **Anthropic Claude**, and **AWS Bedrock** behind a provider abstraction layer to enable model switching based on accuracy, latency, and cost without application code changes.
- Developed an ingestion service to process PDFs and structured content by extracting text, normalizing sections, generating embeddings using Hugging Face Transformers, and updating vector indexes.
- Designed multi-step agent workflows (retrieve, re-rank, generate, validate) to reduce hallucinations and improve grounding for regulated topics, improving “answer-with-citation” **success rate by ~20%** over single-shot prompting.
- Built asynchronous ingestion and indexing using **AWS Lambda** and Step Functions; used AWS Glue and Athena for bulk backfills and audit-style queries during large data loads.
- Implemented Redis caching for frequent queries, embeddings, and repeated LLM calls to reduce response time and control token usage under concurrent traffic.
- Containerized services using Docker and deployed to **AWS EKS** with autoscaling and rolling deployments to support peak internal usage, maintaining **~99.9%** service availability during business hours.
- Implemented observability using **CloudWatch**, **Prometheus**, **Grafana**, and **ELK** to monitor LLM latency, token usage, retrieval hit rate, and error patterns, reducing **MTTR by ~25%** through improved tracing and dashboards.
- Implemented security controls using IAM, KMS, Secrets Manager, OAuth2/JWT, and VPC isolation to protect sensitive data, embeddings, and model outputs.
- Added automated tests for ingestion, retrieval, and prompt templates to validate end-to-end GenAI flows before release.

Environment: Python, FastAPI, GenAI/LLMs, RAG, LangChain, Prompt Engineering, Multi-step Agent Workflows, OpenAI GPT-4, Anthropic Claude, AWS Bedrock, Hugging Face Transformers (Embeddings),

Document Processing (PDF), FAISS, Pinecone, AWS Lambda, AWS Step Functions, AWS Glue, AWS Athena, Redis, Docker, Kubernetes (AWS EKS), Amazon CloudWatch, Prometheus, Grafana, ELK Stack, AWS IAM, AWS KMS, AWS Secrets Manager, OAuth2, JWT, VPC, Automated Testing.

Software Engineer

December 2021-August 2023

Medica, US

Designed and supported Python-based backend and data platforms to enable internal business operations, reporting, and analytics workflows. Built REST APIs and data ingestion pipelines to process data from databases, files, and third-party APIs, performing transformations and analysis using Pandas/NumPy and optimized SQL. Later contributed to AI-assisted capabilities by integrating OpenAI APIs and an early RAG-style workflow with prompt validation and safe fallbacks, delivering features in an Agile/Scrum environment.

- Worked as a **Python Developer** on backend and data platforms supporting internal business operations, reporting, and analytics use cases.
- Built and maintained Python-based **REST APIs** to expose processed data to internal applications, handling validation, error handling, and business logic.
- Designed and implemented data ingestion pipelines in Python to pull data from relational databases, flat files, and third-party APIs on scheduled and event-driven intervals.
- Used **Pandas and NumPy** extensively for real-time data cleaning, transformation, aggregations, and time-based analysis required for reporting and downstream systems.
- Wrote and optimized SQL queries (joins, indexes, aggregations) on relational databases to support both transactional workflows and analytical dashboards.
- Implemented batch and asynchronous processing jobs to handle large data volumes without impacting live production systems, improving pipeline reliability to **~99%** successful runs.
- Deployed **Python services on Linux environments**, using shell scripts for job scheduling, log rotation, monitoring checks, and operational automation, cutting routine ops time by **~20–30%**.
- Supported production issues by debugging data inconsistencies, pipeline failures, and API performance issues, working closely with QA and business users.
- After **GenAI** adoption began, contributed to internal AI-assisted features, integrating Python services with OpenAI APIs to generate summaries and insights from structured datasets.
- Helped implement an early **RAG-style workflow**, where documents and reports were processed in Python, embedded, and queried to provide context-aware responses for internal users, improving relevance of retrieved context in UAT by **~15–20%**.
- Added prompt templates, response validation, and fallback handling to ensure GenAI outputs were safe, relevant, and usable within business applications.
- Collaborated in an Agile/Scrum environment, participating in sprint planning, code reviews, and incremental releases as systems evolved toward AI-assisted workflows.

Environment: Python, REST APIs, Data Pipelines/ETL, Pandas, NumPy, SQL, Relational Databases, Linux,

Shell Scripting (Bash), Batch Processing, Asynchronous Jobs, OpenAI APIs, RAG (early workflow), Prompt Templates, Response Validation/Fallback Handling, Agile/Scrum, QA Collaboration, API Debugging/Production Support.

Software Engineer

December 2019–November 2021

First Interstate BancSystem

Built and supported full-stack web applications using Python/Django and J2EE to deliver end-to-end business features. Developed client-facing pages and modern UI components using HTML/JSP/XML, React, JavaScript, and CSS with dynamic updates via JSON. Improved production stability by troubleshooting critical issues, managing defects in JIRA, and collaborating in an Agile team environment. Automated CI pipelines with Git/Jenkins/MySQL/Bash and standardized Splunk deployments on AWS across UNIX and Windows systems.

- Built client-facing pages using **HTML, JSP, and XML** to support user interactions and application workflows.
- Developed full-stack modules using **Python and Django**, delivering end-to-end features across front end and back end.
- Created modern UI components using **React.js, JavaScript, CSS, Node.js**, and Underscore.js, managing code with Git.
- Troubleshot production issues, fixed defects, and deployed multiple bug fixes for two business-critical applications used by customers and internal support teams.
- Collaborated in an Agile team environment, participating in sprint activities, code reviews, and iterative deliveries.
- Used **Python and Django** for XML processing, data exchange, graphics generation, and backend business logic implementation.
- Implemented dynamic UI updates using **JavaScript and JSON** to refresh partial web page content without full reloads.
- Managed Agile delivery using JIRA for defect tracking and sprint coordination while troubleshooting and resolving functionality, performance, and UI issues across JavaScript, jQuery, HTML5/CSS3, AngularJS, and React.js.
- Migrated and standardized **Splunk deployments on AWS**, configuring forwarders and maintaining consistent setups across UNIX and Windows environments.
- Automated CI workflows using **Git, Jenkins, MySQL, and Bash**, and integrated REST-based microservices to deliver accessible, cross-browser UI pages for dealership platforms.
- Used Python libraries such as **Matplotlib** (visualizations), Pandas (dataframes), Pickle (serialization), PyMongo (MongoDB), PySpark (big data), PySide (UI), and MySQL connectors for database integration.

Tech Stack: HTML, JSP, XML, Servlets, JavaBeans, Python, Django, JavaScript, CSS, React.js, Node.js, Underscore.js, Git, Agile/Scrum, JIRA, JSON, jQuery, HTML5, CSS3, AngularJS, Splunk, AWS, UNIX, Linux,

Windows, Jenkins, MySQL, Bash, REST APIs, Microservices, Pandas, Matplotlib, Pickle, PyMongo, MongoDB, PySpark, PySide.

Software Developer

May 2018-November 2019

Ericsson

Developed and deployed an end-to-end web application using **Python** and **Django**, delivering scalable front-end and back-end modules aligned with enterprise standards. Focused on requirement analysis, database-driven data management with **MySQL**, real-time executive dashboards using the ELK stack, Graphite, and Redis, along with automation scripts for XML parsing and Jira database migration, enabling reliable reporting, system integrations, and operational visibility on AWS/Linux environments.

- Analyzed business needs and documented functional/non-functional requirements, designing the solution architecture aligned with J2EE standards.
- Designed, developed, tested, deployed, and maintained web applications end-to-end following **SDLC practices**.
- Built complete front-end and back-end modules using Python and the Django web framework.
- Developed responsive UI components using **HTML, CSS, JavaScript**, AJAX, and jQuery for interactive user experiences.
- Implemented **JSON-based RESTful web services** for seamless integration between client and server modules.
- Designed and developed a data management layer using MySQL, writing complex queries to extract, store, and transform data.
- Automated ingestion workflows by writing **Python scripts to parse XML** documents and load processed data into databases.
- Created executive real-time dashboards using **Elasticsearch, Logstash, Kibana**, Graphite, and Redis for monitoring and insights.
- Generated **dynamic configuration**/property lists for multiple applications using reusable Python utilities.
- Supported **Jira** database migration by building Python scripts to transfer, match fields, validate records, and verify data integrity.
- Enforced Python best practices across cross-functional, multi-platform systems and provided guidance on scalable long-term architecture decisions.
- Used Pandas for time-series and tabular data manipulation, enabling efficient timestamp-based retrieval and reporting.
- Followed a test-driven approach and implemented unit tests using the Python Unit Test (PyUnit) framework.
- Used PyQuery for DOM selection and HTML parsing during data extraction and automation tasks.
- Deployed **AWS-hosted web applications** on Linux servers, automating setup and deployment steps using Bash scripting.

Environment: Python, Scikit-learn, TensorFlow, Flask, Pandas, NumPy, Matplotlib, Seaborn, Pickle, Joblib, PyTest, AWS EC2, HTML, CSS, Bootstrap, REST APIs, Agile/Scrum.

Education

Master's information systems and technology
University of North Texas, USA

Bachelors in Electronics and Communication Engineering
Osmania University, India